Questions by topic: Forest resources & deforestation - Set 1 (70 marks)

Answer all questions in the spaces provided.

0 1 . 1 Describe the resources obtained from forests. Give **one** example for each category.

[5 marks]

Timber .....

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Fuel .....

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Food .....

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Fibres .....

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Medicines .....

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0 1 . 2 Explain why forests are an important carbon reservoir and how they can be used to counteract climate change.

[3 marks]

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0 1 . 3 Explain how forests regulate the hydrological cycle and local climate.

[2 marks]

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0 2 Table 1 compares an indigenous mixed broadleaf woodland with a Sitka spruce plantation of the same area in Scotland.

Table 1

Feature	Mixed indigenous woodland	Sitka spruce plantation
Tree species	12	1
Age structure	All ages, dead wood present	All planted in the same year
Planting density	Natural, variable	Close, 2 m spacing
Timber yield / m^3 ha^{-1} yr^{-1}	4	16
Bird species	38	9
Ground flora species	60	4

0 2 . 1 Use Table 1 to explain why the plantation has a much higher timber yield but far lower biodiversity.

[6 marks]

0 2 . 2 Explain why commercial plantations are felled before the trees reach full size.

[2 marks]

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0 2 . 3 Explain why there is little demand today for coppiced or pollarded wood.

[2 marks]

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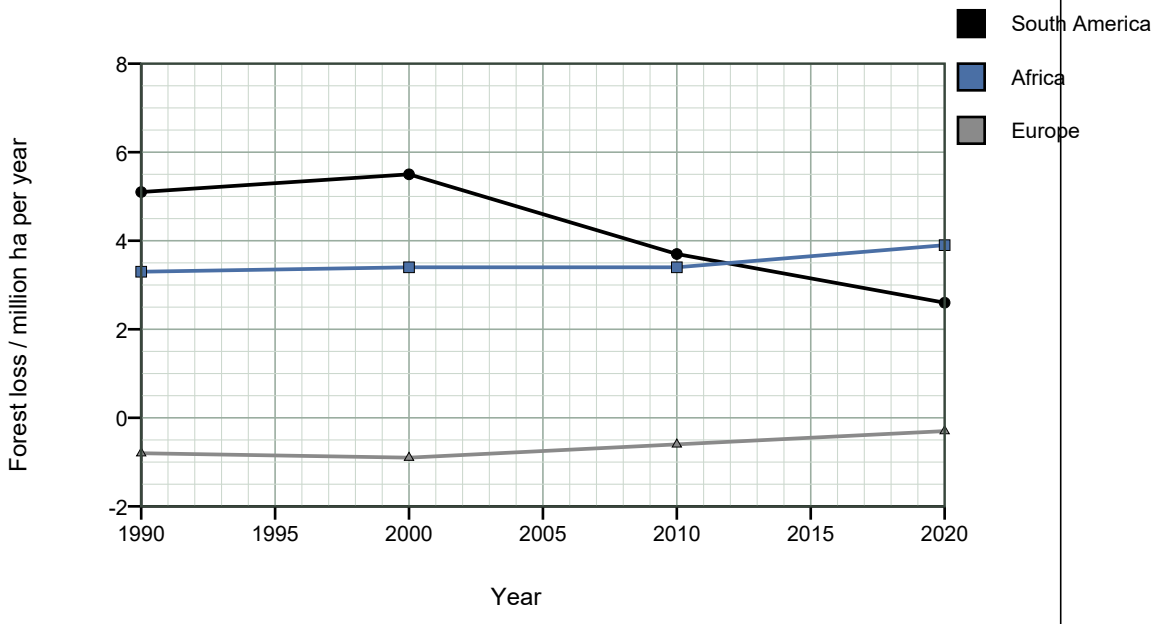
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0 3 Figure 1 shows the area of forest lost each year in three regions between 1990 and 2020.

Figure 1



0 3 . 1 Describe the trends shown in Figure 1.

[3 marks]

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0 3 . 2 Explain the main causes of forest loss in South America and Africa.

[4 marks]

0 3 . 3 Explain why roads and railways cause more forest loss than the area they occupy.

[2 marks]

0 3 . 4 Explain why softwood from Scandinavia and Canada is generally produced more sustainably than tropical hardwood.

[1 mark]

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0 4 . 1 Explain the effects of deforestation on soil.

[4 marks]

0 4 . 2 Explain how deforestation affects hydrology and climate.

[4 marks]

0 4 . 3 Give **two** ecological impacts of deforestation other than the direct loss of species.

[2 marks]

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2 .....

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0 5 The Rio Bravo Conservation and Management Area in Belize exploits mahogany under a sustainable management plan while conserving a rainforest of over 200 tree, 400 bird and 70 mammal species.

0 5 . 1 Describe **five** features of the timber harvesting at Rio Bravo that make it sustainable.

[5 marks]

1 .....

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2 .....

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0 5 . 2 Explain why involving local communities is essential to the success of the reserve.

[3 marks]

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0 5 . 3 Explain why selective logging is better for wildlife than clear-felling, and give **one** disadvantage.

[2 marks]

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0 6 . 1 Which of the following features of a plantation would most increase its wildlife value? Tick one box.

[1 mark]

A Close planting of a single non-indigenous species ☐

B Mixed indigenous species with a mixed age structure ☐

C Clear-felling on a 40-year rotation ☐

D Removal of dead wood ☐

0 6 . 2 Describe **four** features of sustainably managed forests and explain how each benefits biodiversity or productivity.

[4 marks]

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0 7 The Rio Bravo Conservation and Management Area in Belize is a largely undamaged rainforest with over 200 species of tree, 400 species of bird and 70 species of mammal. It is managed by a Belizean NGO to conserve wildlife, produce resources, sequester carbon and benefit the local economy.

0 7 . 1 Explain why creating the reserve could have threatened local communities, and describe how the project avoids this.

[3 marks]

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0 7 . 2 Explain **four** of the rules followed when trees are selected and felled that keep the exploitation sustainable.

[4 marks]

0 7 . 3 Each felled tree is given a unique identification tag. Explain how this supports Forest Stewardship Council accreditation.

[2 marks]

0 7 . 4 Describe **three** ways the reserve raises money without damaging the forest.

[3 marks]

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0 7 . 5 Explain why selective logging of this kind causes much less ecological damage than clear felling.

[3 marks]

15

END OF QUESTIONS